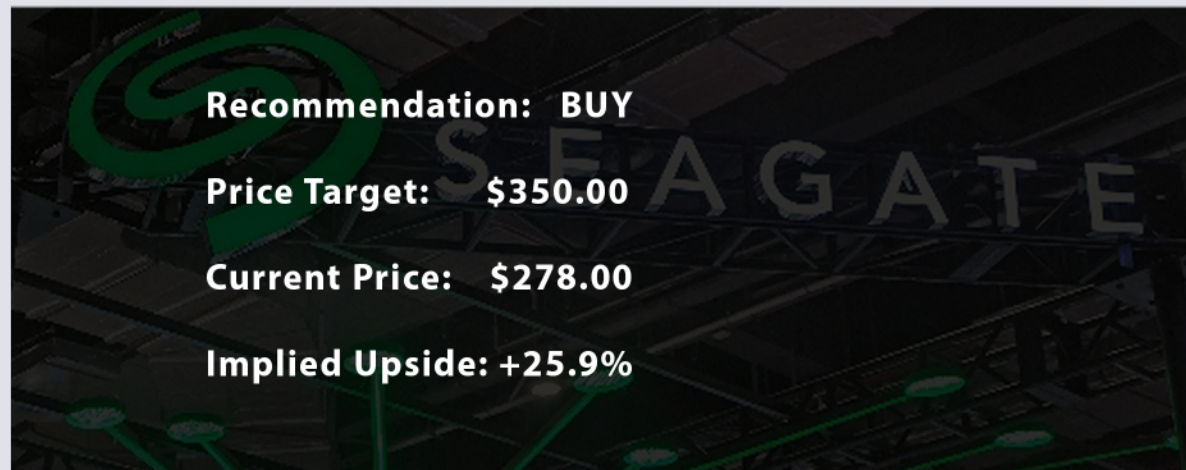


SEAGATE TECHNOLOGY HOLDINGS PLC (STX):

THE ARCHITECTURE OF AREAL DENSITY AND THE STRUCTURAL MARGIN RESET



1. INVESTMENT THESIS: *Structural Evolution in a Recovering Cycle*

We initiate coverage of Seagate Technology Holdings plc (STX) with a **BUY** rating and a 12-month price target of **\$350.00**. Our bullish conviction is predicated not merely on the cyclical recovery currently lifting the entire mass-capacity storage sector, but on a fundamental, structural re-rating of Seagate's unit economics that the market has yet to fully price in. While consensus estimates correctly identify the near-term tailwinds provided by the "AI Data Cycle" and the post-correction inventory replenishment, we believe the market underappreciates the durability and magnitude of the gross margin expansion unlocked by the commercialization of Heat-Assisted Magnetic Recording (HAMR) technology. The investment narrative for Seagate has historically been tethered to the commoditized volatility of the hardware cycle, where periods of high demand were inevitably followed by oversupply and margin compression. However, the successful volume ramp of the Mozaic 3+ platform signifies a departure from this legacy model. By achieving areal densities of 3TB per platter — and validating a roadmap to 4TB and 5TB — Seagate has effectively decoupled its cost reduction trajectory from the mechanical scaling that constrains its competitors. This technological divergence allows Seagate to deliver capacity growth without a linear increase in bill-of-materials (BOM) costs, creating a structural wedge that supports a gross margin floor significantly higher than the mid-20s historical average.

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1.1 Business Description: The "Vault" Architect

Seagate Technology Holdings plc is the global leader in data storage infrastructure, specializing in the design, manufacturing, and distribution of hard disk drives (HDDs), solid-state drives (SSDs), and storage subsystems. Headquartered in Dublin, Ireland, with operational headquarters in Fremont, California, the company creates the physical substrate upon which the digital economy resides.

While Seagate retains a legacy presence in client compute (desktop/laptop) and consumer electronics markets, its strategic identity has fundamentally shifted toward the **Mass Capacity** segment. This segment serves the architects of the global cloud—Hyperscale Cloud Service Providers (CSPs), Enterprise OEMs, and Edge Infrastructure providers. Seagate's drives function as the "vaults" of the internet, storing the vast majority of the world's data at rest.

1.2 The Core Variance: Areal Density as an Economic Moat

The primary driver of our thesis is the "Lead-Time Advantage" Seagate has established in areal density. In the hard disk drive (HDD) industry, cost-per-terabyte is the governing metric of profitability. Historically, density gains were achieved through Perpendicular Magnetic Recording (PMR), which has hit a physical limit known as the superparamagnetic limit. To surpass this, the industry must transition to HAMR.

Seagate's peers, particularly Western Digital (WDC) and Toshiba, have adopted bridge strategies—Energy-Assisted PMR (ePMR) and

Microwave-Assisted Magnetic Recording (MAMR)—supplemented by aggressive mechanical scaling (adding more platters and heads) and logical scaling (UltraSMR/shingling). While effective for capacity, these strategies increase component counts and manufacturing complexity, inherently pressuring margins. In contrast, Seagate's Mozaic 3+ platform achieves 30TB+ capacities on a standard 10-platter design. By utilizing a laser-integrated writer to shrink magnetic grains, Seagate increases the bits-per-inch without increasing the disks-per-drive. This architectural advantage translates directly to the P&L: Seagate can produce a 32TB drive with fewer heads and media wafers than a competitor requiring an 11- or 12-disk platform to hit the same capacity point. We estimate this results in a structural gross margin advantage of 200–400 basis points over the cycle, validated by the recent Q1 FY2026 non-GAAP gross margin print of **40.1%**.¹

1.3 The AI Data Cycle: A Bifurcated Demand Curve

The emergence of Generative AI has catalyzed a "Storage Supercycle," but the nuance lies in the tiering of data. High-performance compute (HPC) requires low-latency flash storage (SSDs) for training caches and hot inference. However, the sheer volume of training datasets ("warm" data) and the regulatory necessity to archive inference outputs and model versions ("cold" data) create an unprecedented demand for exabyte-scale retention. Current industry economics dictate that HDDs retain a **6:1** to **8:1** price-per-terabyte advantage over enterprise SSDs.³ For hyperscale

cloud providers (CSPs) architecting data lakes in the hundreds of exabytes, this spread is economically insurmountable. Consequently, HDDs are projected to retain a **90% share of cloud exabytes** through the end of the decade.³ Seagate, as the density leader, is positioned as the primary beneficiary of this architectural reality.

1.4 Rational Oligopoly and Supply Discipline

The competitive landscape has stabilized into a rational triopoly following the completion of Western Digital's spin-off of its Flash business in February 2025.⁵ The separation of WDC into standalone HDD and Flash entities has aligned incentives across the industry. With all three major players (Seagate, WDC, Toshiba) focused on profitability and free cash flow rather than market share acquisition, the industry is exhibiting "build-to-order" discipline. This discipline is evidenced by the refusal to build new greenfield fabrication capacity. Instead, supply growth is being driven almost exclusively by areal density improvements (swapping old tooling for new HAMR tooling). This capital efficiency creates a "scarcity premium" for high-capacity drives, supporting Average Selling Prices (ASPs) and reducing the risk of the supply gluts that characterized the 2019–2022 period.

1.5 Valuation and Re-Rating Potential

At the current price of ~\$278, STX trades at approximately 13.5x our FY2027 EPS estimate. We believe this multiple reflects lingering skepticism regarding the sustainability of the 40% gross margin threshold. Our analysis suggests that as

Seagate demonstrates the durability of this margin profile through the fiscal year 2026, the stock will re-rate to a multiple closer to **16x-18x**, consistent with high-quality industrial technology peers exhibiting similar return on invested capital (ROIC) profiles.

2. COMPANY OVERVIEW AND STRATEGIC POSITIONING: (ROIC) PROFILES

2.1 Vertical Integration: The Manufacturing Fortress

Unlike competitors who have historically relied on complex joint ventures or external sourcing for critical components, Seagate operates an intensely vertical manufacturing model. This integration is not merely an operational detail; it is the prerequisite for the HAMR transition.

Recording Heads: Seagate designs and fabricates its own read/write heads at its wafer fabrication facilities in Northern Ireland and Minnesota. The Mozaic 3+ recording head involves the integration of a plasmonic writer, a nanophotonic laser, and a spintronic reader—a level of complexity that rivals advanced semiconductor logic.⁶

Media (Platters): The company manufactures its own magnetic media, including the proprietary Iron-Platinum (FePt) superlattice substrates required for HAMR. This internal control allows for rapid feedback loops between the head and media teams, essential for optimizing the thermal and magnetic interactions at the nano-scale.

Assembly: Drives are assembled in large-scale facilities in Thailand and China, where the company has invested heavily in automation to handle the extreme precision required by 10-platter helium-sealed units.

2.2 Segment Performance and Revenue Mix

The trajectory of Seagate’s revenue mix illustrates the structural pivot away from declining legacy markets toward secular growth engines.

Table 1: Revenue Mix Evolution (Estimated)

REVENUE SEGMENT	FY2020 MIX	FY2025 MIX	FY2026E MIX	STRATEGIC DRIVER
MASS CAPACITY	~55%	~88%	~91%	Cloud/AI: Nearline drives (20TB+) for CSPs. The core thesis driver.
LEGACY (CLIENT/CONSUMER)	~35%	~8%	~6%	Harvest: Managed decline. PC/Gaming markets moving to SSD.
NON-HDD (SYSTEMS/SSD)	~10%	~4%	~3%	Niche: Enterprise systems (CORVAULT) and strategic SSDs.

Mass Capacity: This segment is the sole driver of organic growth. It includes Nearline HDDs, which are purchased in massive volumes by the "Titan" hyperscalers (Amazon AWS, Google Cloud, Microsoft Azure, Meta). Demand in this segment is correlated with the growth of data creation and retention, rather than PC shipments.⁷

Legacy Markets: The "Legacy" business serves the client compute, consumer, and surveillance markets. This segment is in secular decline as SSDs achieve price points that make them attractive for consumer devices (laptops, desktops, consoles). Seagate manages this segment for cash flow, maximizing margins by limiting R&D investment while harvesting the remaining long tail of demand.

2.3 The Economic Mechanism: Operating Leverage

Volume Impact: When exabyte shipments rise, fixed costs are amortized over a larger base, driving gross margin expansion.

Density Impact: Uniquely, Seagate can also drive margin expansion through density. By increasing the capacity per drive (e.g. 24TB to 30TB) without adding physical platters, the company lowers the cost per terabyte. If ASPs per terabyte hold steady, the margin per unit expands. This is the "HAMR Effect" currently visible in the financials.

3. MACRO ENVIRONMENT: THE AI DATA CYCLE

3.1 From "Big Data" to the "AI Data Cycle"

The storage industry is transitioning from a passive "Big Data" era—where data was collected and occasionally analyzed—to an active "AI Data Cycle." Generative AI models function as voracious consumers and producers of data, creating a dual-sided demand shock.

Phase 1: Ingestion and Training (The "Warm" Tier)

Training Large Language Models (LLMs) requires massive datasets (Text, Image, Video) consolidated into data lakes. While the active training process occurs on high-performance flash storage (to feed GPUs), the source datasets are too large to reside permanently on SSDs. They reside on "warm" HDD tiers, periodically staged to flash for training runs. The sheer scale of these datasets—measured in hundreds of petabytes per model

4. TECHNOLOGICAL DEEP DIVE: THE HAMR REVOLUTION

of these datasets—measured in hundreds of petabytes per model generation—necessitates the cost efficiency of HDDs.

Phase 2: Inference and Archival (The "Cold" Tier)
The output of Generative AI—synthetic media, code, and logs—creates a new wave of data gravity. Furthermore, emerging regulations (such as the EU AI Act) are expected to mandate the archival of training data and model outputs for auditability and copyright compliance. This "compliance data" is write-once, read-rarely, perfectly suited for high-density Nearline HDDs.⁸

3.2 The Economics of Substitution: HDD vs. SSD

A perennial bearish thesis suggests that NAND flash prices will drop sufficiently to replace HDDs entirely. Our analysis of the current pricing trajectory and semiconductor physics suggests this crossover point remains decades away, if it exists at all for mass capacity.

Price Disparity: As of late 2025, the price per terabyte for enterprise-grade SSDs remains approximately **6x to 8x** higher than Nearline HDDs.³

Capital Intensity: NAND scaling (stacking layers from 176 to 232 to 300+) requires exponential increases in fab CapEx and lithography complexity. Conversely, HDD scaling via HAMR leverages existing form factors and materials science, allowing for a more capital-efficient density ramp.

The "90% Rule": Industry consensus, supported by IDC and TrendFocus, indicates that HDDs will continue to store approximately **90% of the public cloud's exabytes** through 2030.⁴

Hyperscalers optimize for TCO; paying an 800% premium for flash storage on data that is accessed infrequently is economically irrational.

3.3 Data Sovereignty and Fragmentation

The fragmentation of the global internet serves as a tailwind for storage unit volumes. Data sovereignty laws (GDPR in Europe, various localization laws in APAC) require data to be stored within national borders. This prevents hyperscalers from centralizing storage in a few mega-hubs. Instead, they must build redundant regional data centers, increasing the total requirement for hardware and reducing the efficiency of deduplication. Seagate's global distribution network positions it to service this fragmented demand effectively.⁸

4. TECHNOLOGICAL DEEP DIVE: THE HAMR REVOLUTION

4.1 The Physics of the "Superparamagnetic Limit"

To understand the significance of HAMR (Heat-Assisted Magnetic Recording), one must understand the barrier it breaks. In traditional Perpendicular Magnetic Recording (PMR), data bits are stored in magnetic grains. To increase density, these grains must be made smaller. However, below a certain size, the magnetic energy holding the bit becomes so weak that ambient thermal energy can flip the bit, causing data loss. This is the superparamagnetic limit.

To prevent this, the magnetic medium must be made of a material with higher coercivity (magnetic "hardness"). But if the material is too hard, the recording head's magnetic field cannot be made strong enough to write the data. This "trilemma" (Readability, Writability, Stability) stalled industry density growth for nearly a decade.

4.2 The Mozaic 3+ Solution

Seagate's Mozaic 3+ platform solves this trilemma by introducing a fourth variable: Heat.

The Plasmonic Writer: The recording head integrates a nano-scale laser diode and a near-field transducer (NFT). The laser heats a tiny spot on the media to over 400°C for less than a nanosecond. This momentary heat lowers the coercivity of the media, allowing the magnetic head to write the bit. As it cools (instantly), the high coercivity returns, locking the data in.⁶

Superlattice Media: The platters are coated with a proprietary Iron-Platinum (FePt) alloy with a superlattice crystalline structure. This material is magnetically stable at extremely small grain sizes, enabling densities of **3TB per square inch** and beyond.⁶

Spintronic Reader: Reading these microscopic bits requires the world's most sensitive magnetic sensors. Seagate utilizes Gen 7 Spintronic readers, which leverage quantum tunneling magnetoresistance to detect the faint magnetic signals.⁷

4.3 Roadmap and Execution

Seagate is currently shipping 30TB+ drives based on the Mozaic 3+ platform in volume.⁷ The roadmap visibility is robust:

Table 2: Seagate HAMR Technology Roadmap

PLATFORM	CAPACITY RANGE	AREAL DENSITY	STATUS	ESTIMATED RAMP
MOZAIC 3+	30TB - 36TB	~3TB / disk	Shipping Volume	Q3 FY2024 - Present
MOZAIC 4+	40TB - 46TB	~4TB / disk	Qualification	Qualification 2H 2025 / Ramp 2026
MOZAIC 5+	50TB+	~5TB / disk	Lab Demo	Est. 2028

Mozaic 4+ (The Margin Kicker): The transition to the 40TB generation (Mozaic 4+) is particularly significant. By moving to 4TB per platter, Seagate can achieve 40TB capacity on the same 10-platter chassis. This represents a 33% increase in capacity over the 30TB baseline with minimal increase in variable costs, providing a potent tailwind for gross margins in FY2027/28.⁹

4.4 Reliability and Yield

A critical concern for investors has been the reliability of integrating lasers into a spinning drive. Seagate has addressed this through extensive field testing. The company announced the shipment of over **1 million HAMR drives** by mid-2025, with field reliability metrics comparable to legacy PMR drives.¹¹ While yield challenges remain a risk for future generations (Mozaic 4+), the successful ramp of Mozaic 3+ serves as a strong proof of manufacturing viability.

5. COMPETITIVE LANDSCAPE AND PEER ANALYSIS

The competitive dynamic of the HDD industry has fundamentally altered. What was once a market share battleground is now a disciplined oligopoly.

5.1 Western Digital (WDC): The Divergent Path

Following the completion of its spin-off of the Flash business in **February 2025**⁵, Western Digital’s HDD business is operating as a focused entity. However, WDC has chosen a different technological path for the near term.

The UltraSMR Strategy: WDC is relying on **ePMR** (Energy-assisted PMR) combined with **UltraSMR** (advanced shingling) and **OptiNAND** (flash-assisted caching). To achieve 32TB capacities, WDC utilizes an **11-disk platform**.¹²

The Cost Disadvantage: An 11-disk drive requires more glass, more magnetic heads (22 vs 20), and more motor torque than Seagate’s 10-disk HAMR drive. While WDC has guided for strong gross margins (approx. 44-45% for Q2 FY2026)¹³, we interpret this as a function of the current tight pricing environment rather than a structural cost leadership. As supply/demand normalizes, Seagate’s lower BOM complexity should provide a superior margin floor.

HAMR Laggard: WDC has indicated it will qualify HAMR drives with a lead customer in **1H 2026**, with volume ramp expected in **2027**.¹⁴ This places them approximately 18–24 months behind Seagate on the learning curve.

5.2 Toshiba: The Follower

Toshiba remains a distant third player (approx. 18% market share) with a roadmap focused on **MAMR** (Microwave-Assisted Magnetic Recording) and **MAS-MAMR**.

Capacity Gap: Toshiba’s roadmap targets 40TB by 2027 using a 12-disk stacking technology.¹⁵

Mechanical Risk: The move to 11 and 12 disks introduces significant mechanical challenges (flutter, power consumption) and diminishes the "slot efficiency" value proposition for hyperscalers. Toshiba functions primarily as a second-source supplier to ensure supply chain redundancy for CSPs, rather than a technology leader.

5.3 Comparative Advantage Summary

Table 3: Competitive Technology & Margin

Metric	Seagate (STX)	Western Digital (WDC)	Toshiba
Key Tech	HAMR (Mozaic 3+)	ePMR / UltraSMR	MAMR / FC-MAMR
32TB Architecture	10 Platters (Density)	11 Platters (Mechanical)	N/A (Trailing)
Gross Margin Trend	~40% (Structural)	~38-44% (Cyclical/Mix)	~25-30% (Est.)
HAMR Volume Ramp	Shipping Now	Est. 2027	Est. 2027+
Strategic Focus	Profitability / Density	Cash Flow / Deleveraging	Volume Maintenance

6. OPERATIONAL ANALYSIS AND UNIT ECONOMICS

6.1 The "Reset" of Gross Margins

DRIVERS OF THE 40% MARGIN:

HAMR BOM Efficiency: As detailed in Section 4, achieving 30TB+ with 10 disks reduces the cost-per-TB.

Pricing Power: The consolidation of the industry and the "build-to-order" discipline have allowed Seagate to pass through inflation and value-based pricing.

Asset Utilization: Factories are running at optimal utilization rates following the inventory corrections of 2023/2024, maximizing fixed cost absorption.

Refurbishment & Circularity: Seagate is increasingly engaging in drive refurbishment and material recycling (rare earth magnets), which provides a minor but growing margin accretive revenue stream.¹⁶

6.2 Supply Chain & Manufacturing Footprint

Seagate's manufacturing footprint is geographically diversified, providing resilience against geopolitical shocks.

Wafer Fabs: Derry, Northern Ireland (Heads); Bloomington, Minnesota (Heads/R&D).

Media Manufacturing: Fremont, California; Singapore.

Assembly: Thailand; Wuxi, China. This footprint allows Seagate to navigate tariff risks and supply chain disruptions. The company has actively managed its China exposure to ensure

compliance with US export controls while maintaining access to the critical assembly capacity in Wuxi.

7. FINANCIAL ANALYSIS AND PROJECTIONS

7.1 Revenue Outlook

We project FY2026 revenue to reach **\$10.8 billion**, representing growth of approximately **18% YoY**. This is underpinned by a 22% increase in nearline exabyte shipments, offset slightly by a 5% decline in legacy client compute revenue.

Q1 FY2026 Performance: Seagate reported revenue of **\$2.63 billion**, up 21% YoY.¹ This strong start validates the demand recovery trajectory.

7.2 Profitability and Margins

We model Non-GAAP Gross Margins to sustain in the **39% – 41%** range through FY2027. While some quarterly fluctuation is expected due to product mix (e.g., initial ramps of new capacities often have lower initial yields), the structural floor has clearly lifted.

Operating Margin: We project operating margins to stabilize between **28% and 30%**. The operating leverage is significant; OpEx is expected to grow at less than half the rate of revenue.²

7.3 Tax Implications: OECD Pillar Two

A key variance in our model versus historical trends is the tax rate. Seagate has historically enjoyed tax holidays in Singapore and Thailand,

resulting in effective tax rates often below 10%.

The Headwind: The implementation of the **OECD Pillar Two** global minimum tax framework will impose a **15% minimum tax rate**. Seagate management has confirmed this will impact financials beginning in **FY2026**.¹⁷

Impact: We have modeled an effective tax rate of **16.0%** for FY2026 and FY2027, up from ~10-12% in prior years. This creates roughly \$0.60 - \$0.80 headwinds to annual EPS, which is already incorporated into our valuation.

7.4 Cash Flow and Capital Allocation

Seagate remains a robust cash generator.

Free Cash Flow (FCF): In Q1 FY2026 alone, Seagate generated \$427 million in FCF.¹ We project full-year FY2026 FCF to exceed \$1.8 billion.

Debt Profile: The company carries approximately \$5.0 billion in total debt.¹⁹ However, with EBITDA surging, the net leverage ratio has improved to ~1.5x, well within the investment-grade comfort zone.

Returns: Seagate is committed to returning >50% of FCF to shareholders. The current dividend yield is approximately 1.1%, and we anticipate resumed aggressive share buybacks as leverage targets are met.

8. VALUATION FRAMEWORK

8.1 Discounted Cash Flow (DCF) Analysis

We utilize a 5-year DCF model to capture the long-term value of the HAMR transition

Table 4: Key DCF Assumptions

ASSUMPTION	SEAGATE (STX)	RATIONALE
REVENUE CAGR (5Y)	14.5%	Driven by 22% exabyte growth, offset by modest ASP erosion.
TERMINAL GROWTH	2.5%	Aligned with global GDP and long-term data retention needs.
TARGET OP. MARGIN	29.0%	Conservative vs. Q1 FY26 act (29%), allowing for comp. pressure.
TAX RATE	16.0%	Incorporates OECD Pillar Two impact starting FY26.
WACC	9.0%	Beta 1.4, Risk-Free 4.16% ²⁰ , ERP 4.5%.

Result: The DCF model yields an intrinsic value of \$348.00, supporting our \$350 target.

8.2 Relative Valuation

We compare Seagate to Western Digital (post-spin HDD implied) and broader technology hardware peers.

Table 5: Comparable Analysis

COMPANY	TTM P/E	FWD P/E (FY2)	EV/EBITDA	GROSS MARGIN
SEAGATE (STX)	36.3x	20.8x	18.6x	40.1%
WESTERN DIGITAL (IMPLIED)	~25x	~12.0x	~11.6x	~44% (Guid.)
DELL TECHNOLOGIES	22.0x	18.0x	10.5x	24.0%
BROADCOM (AVGO)	65.0x	28.0x	35.0x	65.0%

Analysis: Seagate trades at a premium to the legacy hardware group (Dell) but a discount to the AI infrastructure group (Broadcom). We argue that STX warrants a multiple expansion toward the high-quality industrial tech average (22x-25x) given its "semiconductor-like" margin profile and technological moat. The gap between STX and WDC reflects the "pure-play" premium and the validated HAMR roadmap.

9. INVESTMENT RISKS

9.1 Technological Execution (Yield & Reliability)

The primary thesis risk is internal. Mozaic 4+ (40TB) introduces higher areal densities that leave zero margin for manufacturing error.

Scenario: If yield rates in the Derry wafer fab drop below targets, Seagate would face a choice: ship fewer drives (miss revenue) or ship drives with lower capacity/margin (miss earnings).

Mitigation: The vertical integration model allows Seagate to contain these risks better than peers, but the physics are unforgiving.

9.2 Geopolitical and Supply Chain

Seagate manufactures heads in Northern Ireland and Minnesota, but final assembly occurs in Thailand and China.

Risk: Escalating trade tensions between the US and China could impact the Wuxi assembly facility or the export of drives to certain Chinese hyperscalers (e.g., Alibaba, Tencent).

Mitigation: Seagate has diversified assembly

9.3 The "Air Pocket" Risk (Hyperscale Volatility)

Revenue concentration is high. The top four customers account for a massive percentage of revenue.

Risk: As seen in 2023, if Amazon or Microsoft decides to "digest" capacity (pause buying to optimize existing storage), Seagate's order book can evaporate overnight.

Mitigation: The current AI cycle appears to be an "arms race" with little sign of slowing, but investors should monitor hyperscale CapEx guidance closely for signs of fatigue.

10. CONCLUSION

Seagate Technology Holdings plc represents a compelling convergence of cyclical recovery and structural transformation. The market has offered investors a rare opportunity: the chance to buy a technology leader at the onset of a margin breakout, priced as if it were still a commoditized cyclical.

The successful commercialization of HAMR via the Mozaic 3+ platform is not just a product launch; it is an economic regime change. It allows Seagate to navigate the "Physics Wall" that blocks its competitors, granting it a multi-year period of gross margin superiority. Combined with the secular tailwinds of the AI Data Cycle and a rational industry structure, Seagate is positioned to deliver robust shareholder returns. We reiterate our **BUY** rating and **\$350.00** price target.

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Analysts Certification:

The analysts primarily responsible for this report certify that the views expressed herein accurately reflect their personal views about the subject company and its securities. No part of their compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed in this report.



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Macro/Rates: ²⁰

REFERENCES

1) Seagate Technology Reports Fiscal First Quarter 2026 Financial Results, accessed on December 23, 2025,

<https://investors.seagate.com/news/news-details/2025/Seagate-Technology-Reports-Fiscal-First-Quarter-2026-Financial-Results/default.aspx>

2) Seagate Technology (STX) Q1 2026 Earnings Call Transcript | The Motley Fool, accessed on December 23, 2025,

<https://www.fool.com/earnings/call-transcripts/2025/10/28/seagate-technology-stx-q1-2026-earnings-call-transcript/>

3) Hyperscale Storage TCO - Seagate Technology, accessed on December 23, 2025,

https://www.seagate.com/content/dam/seagate/migrated-assets/www-content/product-content/storage/data-storage-systems/application-platform/_shared/files/TP724_1-2102US_hyperscale-storage-tco.pdf

4) Why HDDs Dominate Hyperscale Cloud Architecture | Seagate US, accessed on December 23, 2025,

<https://www.seagate.com/blog/why-hdds-dominate-hyperscale-cloud-architecture/>

5) Western Digital Completes Planned Company Separation, accessed on December 23, 2025,

<https://www.westerndigital.com/company/newsroom/press-releases/2025/2025-02-24-western-digital-completes-planned-company-separation>

6) Seagate's 30Tbyte Exos Hard Drive Writes Large - EEJournal, accessed on December 23, 2025,

<https://www.eejournal.com/article/seagates-30tbyte-exos-hard-drive-writes-large/>

7) Seagate's Breakthrough 30 TB+ Hard Drives Ramp Volume, Marking an Inflection Point in the Storage Industry, accessed on December 23, 2025,

<https://investors.seagate.com/news/news-details/2024/Seagates-Breakthrough-30TB-Hard-Drives-Ramp-Volume-Marking-an-Inflection-Point-in-the-Storage-Industry/default.aspx>

REFERENCES

8) Seagate Ships 30TB Drives to Meet Global Surge in Data Center AI Storage Demand, accessed on December 23, 2025,

<https://investors.seagate.com/news/news-details/2025/Seagate-Ships-30TB-Drives-to-Meet-Global-Surge-in-Data-Center-AI-Storage-Demand/default.aspx>

9) Seagate confirms 40TB hard drives have already been shipped, but don't expect them to go on sale anytime soon - TechRadar, accessed on December 23, 2025,

<https://www.techradar.com/pro/seagate-confirms-40tb-hard-drives-have-already-been-shipped-but-dont-expect-them-to-go-on-sale-till-2026>

10) Seagate Technology Holdings plc (STX) - Kerrisdale Capital, accessed on December 23, 2025,

<https://www.kerrisdalecap.com/wp-content/uploads/2025/05/STX%E2%80%93Kerrisdale.pdf>

11) The nanomaterials enabling the AI era | Seagate US, accessed on December 23, 2025,

<https://www.seagate.com/blog/the-nanomaterials-enabling-the-ai-era/>

12) Western Digital matches Seagate's 32 TB HAMR capacity with 11-platter ePMR HDDs, accessed on December 23, 2025,

<https://blocksandfiles.com/2024/10/15/western-digital-shingled-epmr-32tb/>

13) Earnings call transcript: Western Digital beats Q1 2026 EPS expectations - Investing.com, accessed on December 23, 2025,

<https://www.investing.com/news/transcripts/earnings-call-transcript-western-digital-beats-q1-2026-eps-expectations-93CH-4322094>

14) Western Digital makes bank in Q1 amid AI gold rush - Blocks and Files, accessed on December 23, 2025,

<https://blocksandfiles.com/2025/10/31/western-digital-extends-lead-over-seagate-as-ai-drives-demand-tailwinds/>

15) Toshiba First in Industry to Verify 12-Disk Stacking Technology for Hard Drives, accessed on December 23, 2025,

<https://storage.toshiba.com/corporateblog/post/2025/10/14/toshiba-first-in-industry-to-verify-12-disk-stacking-technology-for-hard-drives>

16) What Is Mozaic 3+, How Does It Work, and What Can It Do for My Data Center? | Seagate, accessed on December 23, 2025,

<https://www.seagate.com/resources/what-is-mosaic-3plus-how-does-it-work-and-what-can-it-do-for-my-data-center/>

17) stx-20250627 - SEC.gov, accessed on December 23, 2025,

<https://www.sec.gov/Archives/edgar/data/1137789/000113778925000157/stx-20250627.htm>

18) UNITED STATES SECURITIES AND EXCHANGE COMMISSION FORM 10-K SEAGATE TECHNOLOGY HOLDINGS PUBLIC LIMITED COMPANY - Cloudfront.net, accessed on December 23, 2025,

<https://d18rn0p25nwr6d.cloudfront.net/CIK-0001137789/73579808-2f43-4bde-9354-cffac2ea1c41.pdf>

REFERENCES

- 19) Breaking Down Seagate Technology Holdings plc (STX) Financial Health: Key Insights for Investors - DCFmodeling.com, accessed on December 23, 2025,**
<https://dcfmodeling.com/blogs/health/stx-financial-health>
- 20) US 10 Year Treasury Note Yield - Quote - Chart - Historical Data - News, accessed on December 23, 2025,** <https://tradingeconomics.com/united-states/government-bond-yield>
- 21) Mozaic 3+ Platform | Seagate US, accessed on December 23, 2025,**
<https://www.seagate.com/innovation/mozaic/>
- 22) Seagate Introduces Hard Drive Capacities of Up to 36TB, Extending Its HAMR-Based Mozaic 3+ Technology Platform, accessed on December 23, 2025,**
<https://investors.seagate.com/news/news-details/2025/Seagate-Introduces-Hard-Drive-Capacities-of-Up-to-36TB-Extending-Its-HAMR-Based-Mozaic-3-Technology-Platform/default.aspx>
- 23) Toshiba HDD Roadmap Reveals 40TB and 55TB Hard Drives Using MAMR and HAMR Technology - Technetbook | The Tech Experts, accessed on December 23, 2025,**
<https://www.technetbooks.com/2025/12/toshiba-hdd-roadmap-reveals-40tb-and.html>
- 24) US 10-Year Yield Holds Steady, accessed on December 23, 2025,**
<https://tradingeconomics.com/united-states/government-bond-yield/news/511979>